Systematic Literature Review: Statistical Artifacts in Digital QEM

Analysis Framework from Chapter 4.3.4.4 applied to 81 collected papers

Updated: April 2026

Contents

1. Introduction

This document contains the automated LLM-assisted literature review of 81 papers collected for the paper "Do We Measure What We Claim? Statistical Artifacts in Quantum Error Mitigation Benchmarks." Ratings and evidence excerpts were generated by a regex scanner (two pattern tiers: strong and weak) followed by an LLM filter (Claude 4.6 Opus) that removed known false positives. The resulting ratings are stored in `data/review_criteria_llm.csv` and serve as an automated reference.

The primary dataset used for all reported statistics in the paper is the human consensus review (`data/review_criteria.csv`), produced by two authors who read all 81 papers with two additional colleagues independently rating 15-paper subsets for reliability verification.

Criterion	Questions to Answer
1. Sample Size	How many shots per circuit? How many circuit instances? How many independent runs?
2. Variance Reporting	Are error bars shown? Confidence intervals? Standard deviations?
3. Statistical Tests	Are comparisons statistically tested? Which tests? p-values reported?
4. Drift Control	How is temporal drift addressed? Interleaved measurements? Recalibration?
5. Overhead Accounting	Is sampling overhead reported? Are comparisons overhead-adjusted?
6. Noise Model	Is the noise model validated? How generalizable are results?
7. Reproducibility	Is code/data available? Are parameters fully specified?
8. Negative Results	Are failure cases reported? When does the method not work?

Table 1: 8-criterion analysis framework for QEM papers.

Ratings: ✓ = adequately addressed, △ = partially addressed, ✗ = not addressed/missing, **n/a** = not applicable.

2. Paper Categories

The papers are organized into the following categories:

1. **QEM Method Papers** — Core techniques (ZNE, PEC, TEM, etc.)
2. **Fundamental Limits & Theory** — Theoretical bounds on QEM performance
3. **Experimental Demonstrations** — Hardware implementations and benchmarks
4. **Benchmarking & Methodology** — Frameworks for evaluating QEM
5. **Reproducibility & Drift** — Studies on temporal stability and reproducibility
6. **Reviews & Surveys** — Comprehensive literature reviews
7. **Adjacent Topics** — QEC, VQA optimization, security, quantum annealing

3. QEM Method Papers

3.1. Tran, Sharma & Temme (2023) — Locality and Error Mitigation of Quantum Circuits [1]

Introduces a locality-aware ("light cone") PEC estimator and ZNE error bound for local observables. The local PEC estimator reduces sampling overhead by orders of magnitude by restricting error cancellation to noise channels inside the observable's causal light cone. ZNE error bound is similarly tightened.

Criterion	Rating	Notes
Sample Size	△	Numerics use $m = 10^6$ samples, 10^4 histogram sets. Circuit counts not systematically varied.
Variance Reporting	✓	Variance of estimators derived analytically; histograms of estimator distributions shown.
Statistical Tests	✗	No formal hypothesis tests or p-values; comparisons are visual.
Drift Control	n/a	Theoretical/numerical study only.
Overhead Accounting	✓	Sampling overhead is a central result; directly compared for standard vs. local PEC.
Noise Model	✓	Pauli-Lindblad noise model with detailed parameterisation; locality assumptions explicitly stated.
Reproducibility	△	Full circuit specs in appendix (IBM Ithaca topology); no code/data repository.
Negative Results	△	Acknowledges ZNE bias cannot be fully removed; no explicit failure-case analysis.

Table 2: Framework analysis: Tran et al. (2023).

3.2. Filippov, Maniscalco & García-Pérez (2024) — Scalability of QEM: from utility to advantage [2]

Thorough analytical comparison of PEC, ZNE, and TEM scaling, including both random and systematic error components. Proves TEM saturates the universal sampling overhead lower bound and identifies circuit-size windows for quantum advantage.

Criterion	Rating	Notes
Sample Size	✓	Total shot budget M explicitly parameterised; contour diagrams show performance vs. circuit size.
Variance Reporting	✓	Random errors derived analytically as $\delta_{\text{random}} \propto \sqrt{\Gamma}/\sqrt{M}$; systematic terms in Table I.
Statistical Tests	✗	Analytical/numerical study; no empirical statistical tests.
Drift Control	✓	Explicitly models noise instability via std. deviation Θ in relative error-rate fluctuations.
Overhead Accounting	✓	Central focus: optimal sampling overheads Γ^* derived and compared.
Noise Model	✓	Sparse Pauli-Lindblad noise; validated by prior experiments; generalizability discussed.
Reproducibility	△	All parameters specified; purely analytical — no code/data repository.
Negative Results	✓	Clearly shows PEC's exponentially worse overhead; identifies circuit-size limits.

Table 3: Framework analysis: Filippov et al. (2024).

3.3. Khan et al. (2024) — Error Mitigation in the NISQ Era [3]

Applied experimental study comparing DD, T-REx, and ZNE on IBM Kyoto and Osaka processors. Reports improvements in expected values and variances but lacks deeper statistical analysis.

Criterion	Rating	Notes
Sample Size	△	Shot counts not explicitly stated; multiple circuit depths but independent runs unclear.
Variance Reporting	△	Variance values reported; no error bars or confidence intervals on plots.
Statistical Tests	✗	No formal statistical tests; comparisons by tabulated values only.
Drift Control	✗	No discussion of temporal drift or recalibration.
Overhead Accounting	✗	Sampling overhead not quantified or discussed.
Noise Model	△	Device error rates reported; no noise model validation or generalisability discussion.
Reproducibility	△	Hardware specs detailed; data "upon inquiry"; no code repository.
Negative Results	△	Notes DD can sometimes worsen results on IBM Kyoto; no systematic failure analysis.

Table 4: Framework analysis: Khan et al. (2024).

3.4. McDonough et al. (2022) — Automated QEM based on Probabilistic Error Reduction [4]

Automated software framework for PER + ZNE, combining gate-set tomography (GST) or Pauli noise tomography (PNT) with noise-scaled quasi-probability distributions. Demonstrated on Rigetti hardware and noisy simulators.

Criterion	Rating	Notes
Sample Size	△	GST uses 550 circuits at 1000 shots each; PER shot counts not fully specified.
Variance Reporting	△	Sampling overhead plots shown; error bars appear on some results but not systematic.
Statistical Tests	✗	No formal statistical tests or p-values.
Drift Control	✗	Not addressed; hardware experiments are point-in-time.
Overhead Accounting	✓	Sampling overhead γ^{ξ} analysed in detail; PER vs. PEC overhead quantified.
Noise Model	✓	Sparse Pauli noise model validated via PNT; GST provides full gate-set characterisation.
Reproducibility	✓	Open-source code on GitHub (Zenodo DOI); uses Mitiq and PyGSTi.
Negative Results	△	Notes GST scales exponentially; PEC overhead acknowledged as prohibitive for deep circuits.

Table 5: Framework analysis: McDonough et al. (2022).

3.5. Zhang et al. (2025) — Demonstrating QEM on Logical Qubits [5]

First experimental demonstration of ZNE on QEC-encoded logical qubits (repetition and surface codes) on superconducting processors. Polynomial extrapolation starting at $r^{\lceil d/2 \rceil}$ effectively mitigates post-correction logical errors.

Criterion	Rating	Notes

Sample Size	✓	43 circuit instances per noise point; multiple code distances (3, 5, 7) and up to 4 QEC rounds.
Variance Reporting	✓	Standard errors shown; bias δ and overhead η scatter plots provided.
Statistical Tests	△	No formal hypothesis tests; simulation compared to experiment with visual overlays.
Drift Control	△	Measurement errors neglected per Supp. Note 5; no explicit recalibration protocol.
Overhead Accounting	✓	Sampling overhead η computed and visualised; bias-overhead trade-off analysed.
Noise Model	✓	Depolarising noise for physical qubits; validated against simulation.
Reproducibility	✓	Data on Zenodo; code on request; processor parameters in Supp. Table 1.
Negative Results	✓	Surface code ZNE less pronounced than repetition code; limitations for qLDPC codes discussed.

Table 6: Framework analysis: Zhang et al. (2025).

3.6. Qin, Chen & Li (2023) — Error Statistics and Scalability of QEM Formulas [6]

Studies how residual error scales with circuit size N. Before mitigation: $O(\varepsilon N)$; after optimised mitigation: $O\left(\varepsilon' \sqrt{N}\right)$. Proposes Importance Clifford Sampling (ICS) for efficient training circuit generation.

Criterion	Rating	Notes
Sample Size	✓	1000 Clifford + 1000 unitary circuits per frame; multiple qubit counts and gate numbers.
Variance Reporting	✓	RMSE is the primary metric; error bars shown; scaling verified numerically.
Statistical Tests	✗	No formal hypothesis tests; scaling verified by visual fit.
Drift Control	n/a	Primarily numerical; IBM hardware validation in supplementary.
Overhead Accounting	✓	N_T training circuits shown to not grow with circuit size; overhead well characterised.
Noise Model	✓	Gate depolarising, composite (dephasing/amplitude damping/ coherent), and gate-dependent.
Reproducibility	△	Parameters fully specified; no code/data repository mentioned.
Negative Results	✓	Notes circuit size still limited by device quality; PEC variance grows exponentially.

Table 7: Framework analysis: Qin et al. (2023).

3.7. Liao et al. (2024) — Machine Learning for Practical QEM [7]

ML-QEM trains classical models (linear regression, random forests, MLP, GNN) to predict noise-free expectation values from noisy data, achieving $\geq 2 \times$ runtime reduction vs. digital ZNE. Validated on up to 100-qubit IBM hardware.

Criterion	Rating	Notes
Sample Size	✓	2000 test circuits for random circuits; 50 training / 250 test on hardware.
Variance Reporting	✓	95% confidence intervals via 1000-fold bootstrap; $\pm$ standard error.

Statistical Tests	△	Bootstrap confidence intervals used; no formal hypothesis tests or p-values.
Drift Control	△	Acknowledges noise "drifts stochastically" but does not explicitly test or control for it.
Overhead Accounting	✓	Runtime overhead reduction quantified: $\geq 2 \times$ overall.
Noise Model	✓	Tests under incoherent, coherent, readout noise; FakeLima + real IBM hardware.
Reproducibility	✓	Code and data on Zenodo; all hyperparameters specified.
Negative Results	✓	ML-QEM fails in extrapolation under coherent noise; RF degrades at high Trotter steps.

Table 8: Framework analysis: Liao et al. (2024).

3.8. Majumdar et al. (2023) — Best Practices for dZNE [8]

Systematic best-practices guide for digital ZNE covering noise amplification, execution (shot allocation), extrapolation (calibration phase diagrams), and composition with readout mitigation and Pauli twirling.

Criterion	Rating	Notes
Sample Size	✓	8000 shots per noise factor; systematic sweep over depths (0-40) and error rates.
Variance Reporting	✓	Shot noise scaling $\propto N_{\text{shots}}^{-1/2}$ demonstrated; std. dev. characterised.
Statistical Tests	✗	Comparisons by RMSE only; no formal hypothesis tests.
Drift Control	△	Discusses gate calibration staleness; recommends twirling but no explicit drift protocol.
Overhead Accounting	✓	Shot overhead scaling explicitly analysed.
Noise Model	✓	Depolarising + coherent errors; calibration phase diagrams show regime-dependent choices.
Reproducibility	✓	Uses open-source Qiskit ZNE prototype; GitHub repository referenced.
Negative Results	✓	Identifies "no fit" regime where dZNE fails; coherent error causes failure without twirling.

Table 9: Framework analysis: Majumdar et al. (2023).

3.9. Wahl et al. (2023) — ZNE on Logical Qubits by Scaling Code Distance [9]

Proposes distance-scaled ZNE (DS-ZNE) for QEC-encoded qubits using code-distance reduction as noise knob. DS-ZNE outperforms unitary-folding ZNE by up to 92% in post-ZNE logical error rate.

Criterion	Rating	Notes
Sample Size	✓	Fixed sampling budget $N_{\text{samples}} = N_{\text{circ}} \times N_{\text{shots}}$; systematic comparison at equal resources.
Variance Reporting	△	Error rates and improvements reported; no explicit error bars on main results.
Statistical Tests	✗	Comparisons by percentage improvement; no formal tests.
Drift Control	n/a	Purely simulation-based.
Overhead Accounting	✓	Explicit overhead comparison via fixed budget; parallelisation benefits quantified.
Noise Model	△	Standard surface code noise model; no hardware validation.

Reproducibility	✓	Uses open-source Mitiq; parameters fully specified.
Negative Results	△	Notes DS-ZNE requires compatible code distances; does not deeply explore failure cases.

Table 10: Framework analysis: Wahl et al. (2023).

3.10. Hour et al. (2024) — Noise-aware Folding for ZNE [10]

Noise-aware folding technique using hardware calibration data to redistribute noise per qubit pair. Achieves 35% improvement in simulators and 31% on real IBM hardware vs. standard folding.

Criterion	Rating	Notes
Sample Size	△	Experiments on simulators and hardware; specific shot counts not clearly stated.
Variance Reporting	△	Improvement percentages reported; error bars not systematically presented.
Statistical Tests	✗	No formal statistical tests; raw accuracy metrics only.
Drift Control	✗	Not addressed; relies on calibration snapshots.
Overhead Accounting	△	Non-uniform folding adjusts counts but total overhead not rigorously quantified.
Noise Model	✓	Leverages calibration data noise models; noise-adaptive compilation integrated.
Reproducibility	△	Algorithm with pseudocode; Qiskit-based but no public repository.
Negative Results	△	Notes challenges scaling to larger circuits.

Table 11: Framework analysis: Hour et al. (2024).

3.11. Jnane et al. (2024) — Quantum Error Mitigated Classical Shadows [11]

Generalizes PEC, ZNE, and symmetry verification to classical shadows. PEC shadows are unbiased estimators of the ideal state with sample complexity identical to conventional shadows up to a multiplicative QEM overhead factor.

Criterion	Rating	Notes
Sample Size	△	Numerical simulations; specific shot counts simulation-dependent.
Variance Reporting	✓	Rigorous sample complexity bounds; variance with overhead factor γ_1 proven.
Statistical Tests	n/a	Primarily theoretical with numerical validation.
Drift Control	△	Acknowledges error model drift; not experimentally tested.
Overhead Accounting	✓	Sampling overhead $\gamma_1 \in O\left(e^{2\xi}\right)$ rigorously derived.
Noise Model	✓	Assumes learned noise model (sparse Pauli-Lindblad or GST); discusses accuracy requirements.
Reproducibility	△	Detailed mathematical framework; no public code/data.
Negative Results	△	Discusses exponential overhead with noisy gates.

Table 12: Framework analysis: Jnane et al. (2024).

3.12. Henao et al. (2023) — Adaptive QEM using Pulse-based Inverse Evolutions [12]

"Adaptive KIK" uses pulse-based inverse evolutions to construct approximate inverse noise channels. Adapts to noise strength, handles moderate-to-strong, time-dependent, and spatially correlated noise. Tested on IBM hardware combined with randomized compiling.

Criterion	Rating	Notes
Sample Size	△	IBM hardware experiments; specific shot counts not prominently stated.
Variance Reporting	△	Error bounds on bias derived analytically; experimental error bars but methodology unclear.
Statistical Tests	✗	No formal statistical tests; visual comparisons.
Drift Control	✓	Explicitly handles time-dependent noise; runs shots over days.
Overhead Accounting	△	Number of circuits independent of system size; total overhead not quantified.
Noise Model	✓	General time-dependent Lindblad framework; handles correlated and coherent noise.
Reproducibility	△	Qiskit module in development; pulse schedules described but not released.
Negative Results	✓	Shows circuit folding produces erroneous results; discusses gate calibration bias.

Table 13: Framework analysis: Henao et al. (2023).

3.13. Filippov et al. (2023) — Scalable Tensor-Network Error Mitigation [13]

TEM constructs a tensor-network inverse of the global noise channel, achieving quadratically lower overhead than PEC ($\gamma_{\text{TEM}} \approx \sqrt{\gamma_{\text{PEC}}}$). Enables mitigation for circuits of twice the depth vs. PEC. Tested up to 100 qubits.

Criterion	Rating	Notes
Sample Size	✓	300 circuits $\times$ 10,000 shots (3×10^6 total); 3×10^5 for 100-qubit experiments.
Variance Reporting	✓	Error bars computed; measurement overhead ratios plotted with theory predictions.
Statistical Tests	✗	No formal tests; visual comparisons against PEC and ZNE.
Drift Control	n/a	Numerical simulation.
Overhead Accounting	✓	$\gamma_{\text{TEM}} \approx \sqrt{\gamma_{\text{PEC}}}$ proven and numerically confirmed.
Noise Model	✓	Sparse Pauli-Lindblad matching van den Berg et al.; noise inversion via MPO.
Reproducibility	△	Algorithm fully described; no public code repository.
Negative Results	△	TEM also has exponentially growing overhead; bond dimension estimation is difficult.

Table 14: Framework analysis: Filippov et al. (2023).

3.14. LaRose et al. (2022) — Error Mitigation Increases Effective Quantum Volume [14]

Demonstrates ZNE increases the effective quantum volume of IBM quantum computers without additional shots, using local unitary folding with Richardson extrapolation.

Criterion	Rating	Notes
Sample Size	✓	$n_c = 500$ circuits, $n_s = 10^4$ total samples; 5 noise-scale factors.
Variance Reporting	✓	2σ error bars via bootstrapping (500 resamples); variance formula in appendix.

Statistical Tests	△	Threshold-crossing condition used but no formal hypothesis test.
Drift Control	✗	No recalibration or interleaving; notes device degradation.
Overhead Accounting	✓	Same total n_s shots for mitigated/unmitigated; overhead zero by design.
Noise Model	△	Model-free; no noise model used or validated.
Reproducibility	✓	Code and data on GitHub.
Negative Results	△	ZNE doesn't always cross 2/3 threshold; method only applies to expectation values.

Table 15: Framework analysis: LaRose et al. (2022).

3.15. Pascuzzi et al. (2022) — Computationally Efficient ZNE [15]

Noise-estimation circuits estimate depolarizing rate p by executing a structurally similar circuit with known output. Combined with readout correction, randomized compiling, and ZNE, achieves near-exact results for 6-qubit Heisenberg model.

Criterion	Rating	Notes
Sample Size	✓	8,192 shots per circuit; 448 randomized compiling instances.
Variance Reporting	✓	Standard error shown as error bars; averaged over 448 instances.
Statistical Tests	✗	Visual comparison to exact solution only.
Drift Control	△	Randomized compiling converts coherent to incoherent errors; no explicit drift monitoring.
Overhead Accounting	△	Three circuit versions required; overhead implicit but not quantified.
Noise Model	✓	Local depolarizing model rigorously analyzed; ZNE improvement proven analytically.
Reproducibility	✓	IBM Q Paris device; circuit diagrams and protocol fully specified.
Negative Results	△	Max error of 0.11 after mitigation; estimation circuit must be structurally similar.

Table 16: Framework analysis: Pascuzzi et al. (2022).

3.16. Liu & Cai (2025) — QEM for Sampling Algorithms [16]

Extends any QEM technique to sampling-based algorithms (beyond expectation values). Cost of error-mitigated output distribution is no greater than estimating a single observable.

Criterion	Rating	Notes
Sample Size	✓	$N_{\text{cir}} = 10^6$ circuit runs in numerical experiments.
Variance Reporting	✓	Variance bounds rigorously derived; TSE computed.
Statistical Tests	△	Hypothesis testing framework proposed for ground state identification.
Drift Control	n/a	Theoretical/numerical work.
Overhead Accounting	✓	Sampling overhead A^2 rigorously derived; same as single observable.
Noise Model	△	Depolarizing model at circuit fault rate; validation limited to one model.
Reproducibility	△	Framework fully specified; no public code/data.
Negative Results	△	Notes negative sample counts can occur; not all algorithms benefit equally.

Table 17: Framework analysis: Liu & Cai (2025).

3.17. Seif et al. (2023) — Shadow Distillation [17]

Combines classical shadows with virtual distillation for state-preparation error mitigation without coherent multi-copy access. Trades circuit complexity for sample complexity. Validated on 5-qubit GHZ state on trapped ions.

Criterion	Rating	Notes
Sample Size	✓	N_U bases and N_S shots systematically varied; MSE averaged over $N_R = 100$ random states.
Variance Reporting	✓	Bootstrap error bars (250 instances); MSE scaling characterised.
Statistical Tests	△	Bootstrap-based error bars but no formal hypothesis tests.
Drift Control	△	Experimental data from prior work; no explicit drift control.
Overhead Accounting	✓	Sample complexity $N_U \sim 2^{0.82 n_q}$ and classical post-processing quantified.
Noise Model	△	Depolarizing for numerics; trapped-ion reveals actual noise sources.
Reproducibility	△	Uses published experimental data; simulation parameters specified; no code shared.
Negative Results	✓	$m = 2$ mitigation has inherent limits; exponential classical processing for $m \to \infty$.

Table 18: Framework analysis: Seif et al. (2023).

3.18. Hicks et al. (2022) — Active Readout-Error Mitigation [18]

Encodes qubits into multi-qubit repetition/Hamming codes immediately before measurement. Analyzes error detection vs. correction trade-offs. Demonstrates 80% readout error reduction on IBMQ Mumbai.

Criterion	Rating	Notes
Sample Size	△	Proof-of-principle on IBMQ Mumbai (5 qubits); shot counts not explicit.
Variance Reporting	△	Variance analysis for encoded vs. unencoded; analytical effective error rates derived.
Statistical Tests	✗	No formal tests reported.
Drift Control	✗	No discussion of drift or recalibration.
Overhead Accounting	✓	Ancilla qubit and CNOT gate overhead quantified; feasibility condition clearly stated.
Noise Model	✓	Symmetric depolarizing noise for CNOTs analytically solved; effective rates for all codes.
Reproducibility	△	IBMQ device named; analytical results derived; no code/data.
Negative Results	✓	Clearly identifies failure when CNOT error exceeds readout error (Fig. 2).

Table 19: Framework analysis: Hicks et al. (2022).

3.19. van den Berg, Minev & Temme (2022) — Model-free Readout-Error Mitigation [19]

Model-free method diagonalizing the measurement channel via random Pauli-X bit flips. Converts arbitrary readout noise into a single multiplicative factor per Pauli observable. Requires no noise model.

Criterion	Rating	Notes
Sample Size	✓	Sample complexity rigorously derived with formal bounds.
Variance Reporting	✓	Shot-noise variance scaling inversely with noise factor explicitly derived.
Statistical Tests	△	Hoeffding's inequality used for probabilistic guarantees.
Drift Control	△	Calibration data reusable but recalibration frequency not discussed.
Overhead Accounting	✓	Overhead characterized by $1/M_{i,i}^2$; compared to quasi-probabilistic methods.
Noise Model	✓	Key feature: no noise model required; arbitrary readout noise handled.
Reproducibility	✓	Complete protocol specified; simulations for up to 12 qubits.
Negative Results	△	For very noisy readout, overhead becomes large; not framed as explicit failure.

Table 20: Framework analysis: van den Berg et al. (2022).

3.20. Krebsbach et al. (2022) — Optimization of Richardson Extrapolation for QEM [20]

In-depth optimization of Richardson extrapolation for ZNE. Shows variance can be precisely controlled by optimizing measurement allocation. Introduces "tilted Chebyshev" nodes that minimise bias for a given variance budget.

Criterion	Rating	Notes
Sample Size	✓	Total budget N_{tot}, effective N_{eff}, per-node N_j all optimized; budgets 10^5-10^{10}.
Variance Reporting	✓	Variance of Richardson estimator derived exactly; overhead Λ^2 precisely controlled.
Statistical Tests	✗	Numerical validation; no formal tests on comparisons.
Drift Control	n/a	Theoretical/numerical study.
Overhead Accounting	✓	Measurement overhead Λ^2 is the central optimization variable.
Noise Model	△	Tested on Markovian and non-Markovian noise; general polynomial expansion assumed.
Reproducibility	✓	Complete protocol; tilted Chebyshev nodes derived analytically.
Negative Results	✓	Linear spacing causes exponential variance growth; diminishing returns identified.

Table 21: Framework analysis: Krebsbach et al. (2022).

3.21. Kim et al. (2025) — Enhanced Extrapolation-Based QEM [21]

Block-level error characterization for structured algorithms (Grover's search). Estimates fidelity of a single repeated core block rather than full circuit. Validated on Aer simulator and IBM Eagle r3.

Criterion	Rating	Notes
Sample Size	✓	$N_{\mathrm{shots}} = 4000$ per run, 10 independent runs.
Variance Reporting	△	10 runs for statistical confidence; error bars not prominently reported.
Statistical Tests	✗	Descriptive comparisons ("over 20% higher"); no formal tests.
Drift Control	✗	No drift discussion on Eagle r3.

Overhead Accounting	△	"Lightweight" approach requiring shallow circuits; exact overhead comparison lacking.
Noise Model	△	Depolarizing on Aer simulator; Eagle r3 physical noise; exponential decay assumed.
Reproducibility	△	Hardware and algorithm specified; no code/data repository.
Negative Results	✓	Standard ZNE fails at high noise (expectation approaches random guessing).

Table 22: Framework analysis: Kim et al. (2025).

3.22. Kiss, Grossi & Roggero (2025) — QEM for Fourier Moment Computation [22]

Echo verification and noise renormalization applied to Hadamard tests for computing Fourier moments. Achieves two orders of magnitude noise reduction on IBM hardware (up to 266 CNOTs, 5 qubits).

Criterion	**Rating**	**Notes**
Sample Size	△	Circuits with 266 CNOTs / 5 qubits; shot counts not detailed.
Variance Reporting	✓	Confidence interval calculation described; error bars on Fourier moments.
Statistical Tests	△	Two orders of magnitude reduction quantified; no formal tests.
Drift Control	△	DD and randomized compiling partially address temporal correlations.
Overhead Accounting	△	Echo verification doubles depth; postselection overhead discussed but not fully quantified.
Noise Model	✓	Depolarizing channel analyzed; noise reduction validated against model predictions.
Reproducibility	△	IBM devices used; circuit construction in appendices; no code/data repository.
Negative Results	△	Limitations of Hadamard tests with non-unitary operators acknowledged.

Table 23: Framework analysis: Kiss et al. (2025).

3.23. Jin et al. (2025) — Noisy PEC and Generalized Physical Implementability [23]

Generalizes physical implementability for PEC with noisy Pauli operations. Proves noiseless cancellation achievable with noisy gates under certain conditions. Derives bounds via diamond norm and logarithmic negativity.

Criterion	**Rating**	**Notes**
Sample Size	n/a	Analytical/theoretical work.
Variance Reporting	n/a	No empirical data.
Statistical Tests	n/a	Mathematical proofs.
Drift Control	n/a	No hardware experiments.
Overhead Accounting	✓	Sampling cost derived via logarithmic negativity.
Noise Model	△	Depolarizing/Pauli-Lindblad models; no hardware validation.
Reproducibility	△	Derivations fully specified; no code/data.
Negative Results	△	Shows limitations when noisy gates are too noisy for cancellation.

Table 24: Framework analysis: Jin et al. (2025).

3.24. van den Berg et al. (2023) — PEC with Sparse Pauli-Lindblad Models [24]

Practical PEC protocol with sparse Pauli-Lindblad noise models on superconducting processors with crosstalk. Efficient model learning (linear parameter scaling) and sampling (linear time). Key IBM infrastructure paper for PEC at scale.

Criterion	Rating	Notes
Sample Size	✓	100 circuit instances × 256 shots per data point; clearly specified.
Variance Reporting	✓	Error bars shown; statistical uncertainty of mitigated values reported.
Statistical Tests	△	Comparisons with unmitigated and ideal results; no formal hypothesis tests.
Drift Control	△	Noise model re-learned periodically; temporal aspects acknowledged.
Overhead Accounting	✓	Sampling overhead quantified; linear scaling demonstrated.
Noise Model	✓	Sparse Pauli-Lindblad validated on hardware; captures crosstalk; linear parameter scaling.
Reproducibility	✓	Protocol fully specified; theoretical guarantees for model learning provided.
Negative Results	△	Limitations at large circuit sizes acknowledged; overhead grows with complexity.

Table 25: Framework analysis: van den Berg et al. (2023).

3.25. Halder et al. (2023) — Development of ZNE Projective Quantum Eigensolver

Develops zero-noise extrapolated projective quantum eigensolver for quantum chemistry applications. Combines ZNE with projective measurement-based techniques on IBM hardware.

Criterion	Rating	Notes
Sample Size	△	Partial specification of shot counts.
Variance Reporting	✓	Error bars reported in figures.
Statistical Evidence	△	Descriptive uncertainty measures only; no inferential testing.
Drift Control	△	Partial drift awareness.
Overhead Accounting	✓	ZNE overhead discussed.
Noise Model	✓	IBM backend noise model used.
Reproducibility	△	Parameters specified; no public code.
Negative Results	✓	Limitations at higher circuit depths reported.

Table 26: Framework analysis: Halder et al. (2023).

3.26. Patil et al. (2023) — QEM Methods Evaluation

Systematic evaluation of multiple QEM methods on IBM hardware for quantum chemistry problems. Compares ZNE, PEC, and measurement error mitigation across different circuit configurations.

Criterion	Rating	Notes
Sample Size	✓	Shot counts clearly specified across configurations.
Variance Reporting	✓	Error bars and statistical spreads reported.
Statistical Evidence	△	Reports uncertainty descriptively but does not use formal tests for method comparisons.
Drift Control	△	Partial awareness of hardware variability.
Overhead Accounting	✓	Overhead per method quantified.

Noise Model	✓	Noise model documentation provided.
Reproducibility	✓	Code and data available.
Negative Results	✓	Failure modes for each method documented.

Table 27: Framework analysis: Patil et al. (2023).

3.27. Weidinger et al. (2023) — QEM Methods Comparison

Comparative study of QEM techniques with focus on practical applicability. Evaluates ZNE variants across different noise regimes on real hardware.

Criterion	**Rating**	**Notes**
Sample Size	△	Partial shot count reporting.
Variance Reporting	✓	Uncertainty measures shown in figures.
Statistical Evidence	△	Descriptive statistics only; no formal inference.
Drift Control	△	Some drift awareness.
Overhead Accounting	△	Partial overhead reporting.
Noise Model	✓	Hardware noise model specified.
Reproducibility	△	Parameters documented; limited code access.
Negative Results	△	Some failure cases noted.

Table 28: Framework analysis: Weidinger et al. (2023).

4. Fundamental Limits & Theory

4.1. Takagi, Endo, Minagawa & Gu (2022) — Fundamental Limits of QEM [25]

Derives universal lower bounds on sampling overhead for any error-mitigation protocol class. Shows: (1) overhead scales exponentially with depth for local depolarizing noise, (2) PEC is optimal for local dephasing. Provides a Carnot-like fundamental limit for QEM.

Criterion	Rating	Notes
Sample Size	n/a	Purely theoretical; bounds on sampling cost M derived.
Variance Reporting	n/a	Bounds on estimator spread; Hoeffding inequality for failure probability.
Statistical Tests	n/a	Provides bounds rather than experimental comparisons.
Drift Control	n/a	Not applicable.
Overhead Accounting	✓	Central contribution: rigorous universal bounds on sampling overhead.
Noise Model	✓	Local depolarizing and dephasing; state distinguishability measures used.
Reproducibility	✓	Fully specified mathematical framework with formal proofs.
Negative Results	✓	The paper *is* about fundamental limits: exponential scaling is unavoidable.

Table 29: Framework analysis: Takagi et al. (2022).

4.2. Takagi, Tajima & Gu (2023) — Universal Sampling Lower Bounds for QEM [26]

Establishes information-theoretic lower bounds covering all existing and future EM protocols (including nonlinear post-processing). Proves exponential sampling cost under depolarizing, stochastic Pauli, and thermal noise.

Criterion	Rating	Notes
Sample Size	✓	The paper's entire focus: necessary sample number N derived as function of accuracy.
Variance Reporting	✓	Bounds in terms of (δ, ε) and (std. dev., bias).
Statistical Tests	n/a	Mathematical proofs.
Drift Control	n/a	No experiments.
Overhead Accounting	✓	Proves $N \geq \Omega(1/(1-\gamma)^{2L})$ is inescapable.
Noise Model	✓	Local depolarizing, stochastic Pauli, global depolarizing, thermal noise.
Reproducibility	△	Proofs in supplemental; fully reproducible from paper.
Negative Results	✓	Universal impossibility bounds.

Table 30: Framework analysis: Takagi et al. (2023).

4.3. Quek et al. (2024) — Exponentially Tighter Bounds on Limitations of QEM [27]

Proves superpolynomial samples needed even at shallow depths. Noise-induced scrambling kicks in at exponentially smaller depths than previously thought. Implications for quantum ML and barren plateaus.

Criterion	Rating	Notes
Sample Size	n/a	Theoretical proof-based paper.

Variance Reporting	n/a	Analytic bounds.
Statistical Tests	n/a	Mathematical proofs.
Drift Control	n/a	Theoretical.
Overhead Accounting	✓	Core contribution: fundamental sample complexity lower bounds.
Noise Model	✓	General framework covering local stochastic noise; broadly applicable.
Reproducibility	✓	Proofs self-contained; framework fully specified.
Negative Results	✓	Entire paper is a fundamental negative result.

Table 31: Framework analysis: Quek et al. (2024).

4.4. Niroula, Gopalakrishnan & Gullans (2025) — Error Mitigation Thresholds [28]

Studies PEC and TEM robustness under imperfect noise characterization. Demonstrates a threshold via Imry-Ma argument: in 1D mitigation fails at $O(1)$ time for any imperfection; threshold exists for $D \geq 2$.

Criterion	**Rating**	**Notes**
Sample Size	△	Numerical simulations; shot counts not always explicit.
Variance Reporting	△	Error bars in figures; limited methodology discussion.
Statistical Tests	✗	No formal tests on comparisons.
Drift Control	n/a	Theoretical/numerical study.
Overhead Accounting	✓	Exponential scaling explicitly derived.
Noise Model	✓	Pauli-Lindblad with systematic imperfection; validates via stat. mech. mapping.
Reproducibility	△	Analytical framework specified; numerical details partially given.
Negative Results	✓	Main result: mitigation fundamentally fails in 1D for any imperfection.

Table 32: Framework analysis: Niroula et al. (2025).

4.5. Jose & Simeone (2022) — Error-Mitigation-Aided Optimization: Convergence Analysis [29]

Analyzes SGD convergence for VQE with and without QEM. Proves quantum noise induces a non-zero error floor, while QEM eliminates it at the cost of increased variance. Derives conditions under which QEM improves iteration complexity.

Criterion	**Rating**	**Notes**
Sample Size	✓	Measurements per iteration, circuit count, and iterations all explicit in convergence bounds.
Variance Reporting	✓	Variance terms derived analytically; bias-variance trade-off is the central analysis.
Statistical Tests	✗	No formal tests on numerical results.
Drift Control	n/a	Simulated noise; no hardware drift.
Overhead Accounting	✓	QEM overhead fully accounted for in convergence bounds.
Noise Model	△	Pauli channel with correlations; standard but not experimentally validated.
Reproducibility	△	Max-cut VQE described but no code/data shared.
Negative Results	✓	Identifies when QEM is not beneficial: requires sufficient noise and circuit count.

Table 33: Framework analysis: Jose & Simeone (2022).

4.6. White (2025) — Managing Errors in Quantum Metrology via QEM (Dissertation) [30]

Theoretical framework for applying ZNE to quantum sensing/metrology. Addresses Markovian noise (Ramsey interferometry) and non-Markovian correlated noise (filter function formalism).

Criterion	**Rating**	**Notes**
Sample Size	△	Shot noise incorporated analytically; no specific experimental counts.
Variance Reporting	✓	Sensitivity (variance-based metric) is the primary figure of merit.
Statistical Tests	✗	Purely theoretical.
Drift Control	✓	Non-Markovian correlated noise explicitly modelled via filter function formalism.
Overhead Accounting	✓	Folding overhead analysed for sensing; overhead-adjusted sensitivity computed.
Noise Model	✓	Phase/amplitude damping (Markovian); arbitrary power spectral density (non-Markovian).
Reproducibility	✓	Full analytical derivations; all models and protocols specified.
Negative Results	✓	ZNE can worsen sensitivity for amplitude damping; effectiveness depends on spectrum shape.

Table 34: Framework analysis: White (2025).

4.7. França & Garcia-Patrón (2022) — Limitations of Optimization Algorithms on Noisy Quantum Devices [31]

Proves that for any quantum circuit of depth d on n qubits subject to local Pauli noise with rate p, the output state converges to the maximally mixed state exponentially: $\mathrm{tr}[O\rho] - \mathrm{tr}[O\rho_{\text{mix}}]$ decays as $e^{-\Omega(pd)}$. This means variational quantum algorithms on noisy hardware face a no-go: once $d = \Omega\left(\frac{1}{p}\right)$ the output is indistinguishable from white noise, making classical brute-force search provably at least as good. The result also implies that quantum error mitigation (QEM) at best delays but cannot eliminate this exponential decay — the "quantum advantage" region shrinks as noise increases. The authors frame this as a "game against the noise," where the noise always wins for deep enough circuits.

Criterion	**Rating**	**Notes**
Sample Size	n/a	Purely theoretical; no numerical experiments.
Variance Reporting	n/a	No empirical data.
Statistical Tests	n/a	Formal mathematical proofs.
Drift Control	n/a	Theoretical analysis; noise is static.
Overhead Accounting	✓	Proves circuit depth threshold $d = \Omega\left(\frac{1}{p}\right)$ beyond which no QEM/ optimization can help — implicitly bounds maximum useful overhead.
Noise Model	✓	Local Pauli noise (depolarising); extends to general local noise with Pauli mixing time bounds. Very general.
Reproducibility	△	Full proofs in appendix; results purely mathematical. No code needed.
Negative Results	✓	The entire paper is a negative result: noisy quantum circuits cannot outperform classical brute-force search once depth exceeds $O\left(\frac{1}{p}\right)$. Directly limits QEM utility.

Table 35: Framework analysis: França & Garcia-Patrón (2022).

4.8. Schultz et al. (2022) — Fundamental Limits of QEM

Derives fundamental performance limits for quantum error mitigation techniques. Establishes information-theoretic bounds on achievable improvement as a function of noise strength and circuit depth.

Criterion	Rating	Notes
Sample Size	✓	Analytical results with clearly specified numerical parameters.
Variance Reporting	✓	Formal variance bounds derived analytically.
Statistical Evidence	△	Mathematical proofs constitute rigorous evidence but in the theoretical-bound sense rather than empirical testing.
Drift Control	✓	Time-independent noise models with explicit theoretical justification.
Overhead Accounting	✓	Overhead bounds are central to the contribution.
Noise Model	✓	Multiple noise models rigorously analysed.
Reproducibility	△	Analytical results reproducible from proofs; no code.
Negative Results	✓	Impossibility results are the paper's main contribution.

Table 36: Framework analysis: Schultz et al. (2022).

5. Experimental Demonstrations

5.1. Saki et al. (2023) — Hypothesis Testing for Error Mitigation [32]

Introduces hypothesis testing for QEM evaluation and an inclusive figure of merit combining resource overhead and mitigation efficiency. Tests 16 error mitigation pipelines (275,640 circuits) on two IBM quantum computers.

Criterion	Rating	Notes
Sample Size	✓	275,640 circuits on two IBM devices; 16 pipelines systematically tested.
Variance Reporting	✓	REM metric with distributional analysis; entropic figure of merit R.
Statistical Tests	✓	One-sample and two-sample hypothesis tests explicitly applied.
Drift Control	△	No explicit drift control; experiments at one time point.
Overhead Accounting	✓	Entropic resource metric R combining shot/circuit/depth overhead.
Noise Model	△	Real hardware noise; no explicit noise model used/validated.
Reproducibility	✓	Full pipeline descriptions; two IBM devices; extensive raw data.
Negative Results	✓	Many pipelines fail hypothesis test; some are worse than no mitigation.

Table 37: Framework analysis: Saki et al. (2023).

5.2. Govia et al. (2025) — Bounding Systematic Error in QEM due to Model Violation [33]

Methodology to upper-bound systematic error in PEC from inaccurate noise models. Bounds computed from Pauli-learning data at no extra cost. Validated on IBM hardware and through simulation.

Criterion	Rating	Notes
Sample Size	△	Infinite-sample limit for systematic error; sampling overhead treated separately.
Variance Reporting	△	Focus on systematic (bias) error; statistical uncertainty acknowledged but not primary.
Statistical Tests	△	Bounds compared to worst observed error; systematic comparison methodology.
Drift Control	✓	Temporal drift explicitly identified as model violation source; methodology extends to drift.
Overhead Accounting	△	No extra experiments needed; PEC overhead deferred to prior work.
Noise Model	✓	Pauli-stochastic channels; five model violation sources cataloged; diamond-distance bounds.
Reproducibility	✓	Named IBM processors; full 4-step protocol; Pauli-Lindblad details.
Negative Results	✓	“Horoscope effect” demonstrated; Cliffords don’t always maximise error.

Table 38: Framework analysis: Govia et al. (2025).

5.3. Maupin et al. (2024) — Error Mitigation on a Trapped Ion Testbed [34]

Three ZNE noise-scaling methods on QSCOUT (trapped-ion) for VQE on HeH^+. Time-stretching and detuning scaling fail; gate identity insertion achieves 96.8% error suppression.

Criterion	Rating	Notes

Sample Size	✓	28,000 samples per experiment; 4-5 noisy energy estimates for extrapolation.
Variance Reporting	✓	Standard error of mean; weighted least-squares regression.
Statistical Tests	△	Chemical accuracy threshold referenced; no formal hypothesis tests.
Drift Control	△	Qubit drift noted after 28,000 samples; optimization minimises drift exposure.
Overhead Accounting	✓	Fixed budget allocation between optimization and extrapolation studied.
Noise Model	✓	Phenomenological noise with depolarizing, crosstalk, coherent; MS gate noise simulated.
Reproducibility	✓	QSCOUT specs in Table I; circuit, Hamiltonian, and optimizer fully specified.
Negative Results	✓	2/3 noise-scaling methods fail; result outside chemical accuracy; non-Markovian issues.

Table 39: Framework analysis: Maupin et al. (2024).

5.4. Kurita et al. (2023) — Synergetic QEM: RC + ZNE for VQE [35]

Combining randomized compiling (RC) and ZNE reduces VQE energy errors from coherent noise by up to two orders of magnitude. RC linearizes noise dependence, enabling ZNE extrapolation.

Criterion	**Rating**	**Notes**
Sample Size	△	35-60 VQE trials with random initial parameters per configuration.
Variance Reporting	✓	Violin plots; median and minimum errors; $1/\sqrt{N_{\text{rand}}}$ scaling confirmed.
Statistical Tests	✗	Visual comparisons via violin plots.
Drift Control	n/a	Numerical simulation only.
Overhead Accounting	△	Identity insertion triples circuit depth; measurement overhead acknowledged.
Noise Model	✓	Four coherent noise models (10^{-3} error rate); no decoherence.
Reproducibility	✓	True-Q, SciPy, OpenFermion; UCC-SD ansatz and noise models fully specified.
Negative Results	✓	RC or ZNE alone insufficient; identity insertion impractical for current NISQ.

Table 40: Framework analysis: Kurita et al. (2023).

5.5. Russo et al. (2023) — Testing Platform-Independent QEM [36]

Evaluates ZNE and PEC across IBM, IonQ, and Rigetti hardware. Defines "improvement factor" normalizing benefit by resources. Finds 1×-7× improvement across all experiments.

Criterion	**Rating**	**Notes**
Sample Size	✓	$N = 10^4$ shots per experiment; $\lvert C \rvert = 4$ random circuits.
Variance Reporting	✓	RMSE defined; standard deviations as error bars.
Statistical Tests	△	Improvement factor as systematic metric; no formal hypothesis tests.
Drift Control	✗	Experiments at different times without interleaving.
Overhead Accounting	✓	Cost factor in improvement factor; same total shots for mitigated/unmitigated.

Noise Model	△	PEC uses local depolarizing; ZNE is model-free; 1% depolarizing simulator.
Reproducibility	✓	All in Mitiq (open source); device errors tabulated; multi-platform.
Negative Results	✓	Some experiments show $\mu < 1$ (worse than no mitigation); IonQ compilation issues.

Table 41: Framework analysis: Russo et al. (2023).

5.6. Kim et al. (2025) — Error Mitigation with Stabilized Noise [37]

Stabilizes qubit-TLS interactions via electrodes on a 6-qubit processor. Compares fixed, optimized, and averaged strategies over 50 hours. Averaged strategy provides most stable PEC results.

Criterion	Rating	Notes
Sample Size	✓	4,096 random circuits $\times$ 32 shots; experiments span 50 hours.
Variance Reporting	✓	300 single-shot measurements; 100 bootstrap instances for PEC.
Statistical Tests	△	Correlation scatter plots; no formal test statistics.
Drift Control	✓	Central contribution: explicitly addresses temporal drift via TLS modulation; 50h monitoring.
Overhead Accounting	✓	Sampling overhead $\gamma = \exp(\sum 2\lambda_k)$ tracked over time.
Noise Model	✓	Sparse Pauli-Lindblad learned and monitored; stability tracked.
Reproducibility	✓	Custom 6-qubit device; all protocols described.
Negative Results	✓	Control experiment shows large fluctuations; averaged strategy has higher overhead.

Table 42: Framework analysis: Kim et al. (2025).

5.7. Weaving et al. (2023) — Benchmarking QEM for HCl Ground State [38]

Benchmarks combinations of measurement-error mitigation, symmetry verification, ZNE, and dual-state purification across 8 IBM Falcon processors for HCl. Systematic compatibility analysis.

Criterion	Rating	Notes
Sample Size	✓	Fixed shot budget; 8 processors with 5-qubit clusters; GHZ up to 27 qubits.
Variance Reporting	✓	Variance via bootstrapping; error bars on fidelity.
Statistical Tests	△	Chemical precision threshold (1.6 mHa) used; no formal hypothesis tests.
Drift Control	△	Hardware specs captured at execution; no interleaving but cross-device provides robustness.
Overhead Accounting	△	Fixed shot budget for fairness; no explicit overhead-adjusted comparison.
Noise Model	✓	Comprehensive hardware characterisation: T1/T2, gate durations, gate/readout errors.
Reproducibility	✓	Named IBM devices; qubit clusters; Hamiltonian tabulated; uses Symmer package.
Negative Results	✓	Sharp fidelity drop beyond 21 qubits; not all combinations improve results.

Table 43: Framework analysis: Weaving et al. (2023).

5.8. Dutkiewicz et al. (2025) — Error Mitigation for Early Fault-Tolerant QPE [39]

Framework for early fault-tolerant algorithms trading between QEC overhead and residual noise. QFT-based QPE (MSQPE) robust to global depolarizing noise. EUMLE extends EM beyond expectation-value estimation.

Criterion	Rating	Notes
Sample Size	✓	Total unitary calls quantified precisely as function of precision ε and noise rate.
Variance Reporting	✓	Std. deviation of phase estimate as target metric; asymptotic normality proven.
Statistical Tests	△	Quantitative cost ratios; no formal hypothesis tests.
Drift Control	n/a	Resource estimation paper; no hardware.
Overhead Accounting	✓	Central focus: EM overhead translated to wall-clock/physical qubit estimates.
Noise Model	✓	Global depolarizing exact; local depolarizing bounded.
Reproducibility	✓	Open-source code; all parameters and QEC architecture assumptions specified.
Negative Results	✓	Diminishing returns beyond $2 \times$ qubit reduction; magic-state costs highlighted.

Table 44: Framework analysis: Dutkiewicz et al. (2025).

5.9. Amin et al. (2023) — Quantum Error Mitigation in Quantum Annealing [40]

Introduces ZNE for quantum annealing on D-Wave Advantage2 (232 qubits). Zero-temperature extrapolation and energy-time rescaling mitigate thermal and control errors.

Criterion	Rating	Notes
Sample Size	△	232-qubit chain; shot counts per data point not explicitly stated.
Variance Reporting	△	Error bars on some plots; methodology not fully specified.
Statistical Tests	✗	Visual comparisons against exact/DMRG solutions only.
Drift Control	✓	Temperature varied systematically; energy-time rescaling accounts for noise scaling.
Overhead Accounting	△	Multiple temperature/energy runs; overhead not quantified.
Noise Model	✓	Open-system theory with Magnus expansion; ohmic environment; validated vs. exact.
Reproducibility	△	Prototype D-Wave hardware; annealing schedules in appendix; no public code/data.
Negative Results	✓	ZNE fails beyond $t_a \gtrsim 200$ ns; temperature extrapolation cannot correct miscalibration.

Table 45: Framework analysis: Amin et al. (2023).

5.10. Kim et al. (2023) — Evidence for the Utility of Quantum Computing Before Fault Tolerance [41]

Landmark experimental demonstration on IBM's 127-qubit Eagle processor (ibm_kyiv/ibm_brisbane). Runs a 60-layer, 2880 two-qubit-gate kicked Ising circuit — the deepest circuit at the time — with ZNE using a sparse Pauli-Lindblad (SPL) noise model for noise amplification. Mitigated expectation values closely match exact Clifford results and DMRG for small bond dimensions, while surpassing classical tensor-network methods (MPS, isoTNS) in the entangled "hard" regime. The paper catalysed intense debate:

Tindall et al. (2024) subsequently showed classical tensor networks could match the results, questioning the “utility” claim. Methodologically notable for its rigorous noise characterisation (3780 two-qubit Pauli error rates learned via cycle benchmarking) but limited drift discussion.

Criterion	Rating	Notes
Sample Size	✓	2×10^4 shots per circuit; multiple circuit depths (1-60 layers); multiple observable strings. Total: $> 10^6$ shots across configurations.
Variance Reporting	✓	Error bars from bootstrapping over shot noise shown on all mitigated expectation values. Systematic error from noise model mismatch discussed.
Statistical Tests	△	No formal hypothesis tests. “Agreement” with Clifford exact results established visually and via χ^2 metric on Clifford subset. No p-values for utility claim.
Drift Control	△	3780 noise parameters re-calibrated daily via cycle benchmarking. No interleaving or explicit drift-correction protocol between calibration and experiment runs.
Overhead Accounting	✓	SPL noise model learning cost (cycle benchmarking) reported. Noise amplification factors $\lambda \in \{1, 1.5, 2, 3\}$. Total shot overhead $\approx 4 \times$ unmitigated. Gate overhead from Pauli twirling reported.
Noise Model	✓	Sparse Pauli-Lindblad model with 3780 learned parameters. Validated on Clifford circuits. Gate-level noise characterisation via cycle benchmarking.
Reproducibility	✓	Device parameters, calibration data, circuit specifications, and noise model details published. Code/data on Zenodo. IBM hardware publicly accessible at time of publication.
Negative Results	✓	Acknowledges the classical simulability debate. Shows mitigated results degrade for very deep circuits. Notes noise model is approximate — does not capture all coherent errors.

Table 46: Framework analysis: Kim et al. (2023).

5.11. Roy et al. (2024) — Simulating Open Quantum Systems Using ZNE [42]

Applies ZNE (via Mitiq) to simulate open quantum system dynamics — amplitude damping and generalised amplitude damping channels — on IBM and Oxford Quantum Circuits (OQC) hardware. Demonstrates that ZNE improves fidelity of open-system simulation across different backends, comparing linear, polynomial, and exponential extrapolation.

Criterion	Rating	Notes
Sample Size	△	Shot counts not always explicit; “default Qiskit shots” used in some experiments. Multiple circuits tested (1-4 qubits).
Variance Reporting	✗	No error bars on mitigated results. No discussion of variance amplification from ZNE extrapolation.
Statistical Tests	✗	No formal tests. Comparisons via fidelity tables and visual plots only.
Drift Control	✗	No drift discussion or correction. Experiments run on cloud hardware without temporal controls.
Overhead Accounting	△	Noise amplification factors (fold multipliers) reported. Total overhead not quantified.
Noise Model	△	Uses gate folding for noise amplification; no independent noise characterisation. Backend noise models not validated.

Reproducibility	✓	Uses Mitiq (open source); circuit specifications provided. IBM/OQC backends specified.
Negative Results	✓	Exponential extrapolation sometimes overshoots. Higher-order polynomial can increase error. Shows ZNE can fail for certain noise regimes.

Table 47: Framework analysis: Roy et al. (2024).

5.12. Shi & Malaney (2023) — Error-Mitigated Quantum Routing on Noisy Devices [43]

Applies ZNE and PEC (both individually and concatenated) to quantum routing protocols on IBM Jakarta (7 qubits). Demonstrates that concatenation of ZNE+PEC provides multiplicative improvement over each method alone for entanglement distribution fidelity.

Criterion	Rating	Notes
Sample Size	✓	8192 shots per circuit; multiple routing protocols (3 different quantum routing circuits). Multiple noise amplification factors tested.
Variance Reporting	△	Error bars on some plots but not all. Variance amplification from PEC overhead γ not quantified.
Statistical Tests	✗	No formal tests. Fidelity improvements reported as percentages.
Drift Control	✗	No drift discussion. IBM Jakarta experiments without temporal controls.
Overhead Accounting	✓	PEC overhead γ reported. ZNE fold factors specified. Concatenation cost discussed.
Noise Model	△	Uses Qiskit's built-in noise model for PEC quasi-probability decomposition. No independent noise characterisation.
Reproducibility	△	Circuit descriptions provided. No code repository. IBM Jakarta since retired.
Negative Results	✓	Shows ZNE alone insufficient for deep routing circuits. PEC overhead grows rapidly. Concatenation has diminishing returns at high noise.

Table 48: Framework analysis: Shi & Malaney (2023).

5.13. Venere et al. (2024) — Characterizing ZNE+QAOA Performance Under Realistic Noise [44]

Short QCE workshop paper (2 pages) applying ZNE to QAOA for Max-Cut on IBM Brisbane noise model (simulated). Characterises ZNE effectiveness as a function of noise strength and circuit depth.

Criterion	Rating	Notes
Sample Size	△	Multiple problem instances; shot counts not explicitly reported.
Variance Reporting	△	Error bars on some plots; limited detail due to 2-page format.
Statistical Tests	✗	No formal tests. Visual comparisons of mitigated vs. unmitigated.
Drift Control	n/a	Simulated noise model only.
Overhead Accounting	✗	Fold factors mentioned but overhead cost not quantified or discussed.
Noise Model	△	IBM Brisbane noise model from Qiskit Aer; no independent validation.
Reproducibility	△	Short format limits methodological detail. Noise model referenced but parameters not fully specified.

Negative Results	△	Notes ZNE effectiveness decreases with depth, but limited analysis due to space constraints.

Table 49: Framework analysis: Venere et al. (2024).

6. Benchmarking & Methodology

6.1. Bultrini et al. (2023) — Unifying and Benchmarking State-of-the-Art QEM [45]

Unifies ZNE, CDR, and Virtual Distillation under the UNITED framework. Benchmarks on realistic trapped-ion noise with random circuits and QAOA for Max-Cut, up to shot budgets of 10^{10}.

Criterion	Rating	Notes
Sample Size	✓	Explicit shot budgets (10^5-10^{10}); multiple circuit sizes and depths.
Variance Reporting	✓	Error bars and confidence intervals across shot budgets.
Statistical Tests	△	Visual comparisons; no formal hypothesis tests.
Drift Control	n/a	Simulated noise.
Overhead Accounting	✓	Shot budget comparisons central; overhead-adjusted performance shown.
Noise Model	✓	Realistic trapped-ion noise; validates across multiple noise regimes.
Reproducibility	△	Parameters specified; code availability not explicitly mentioned.
Negative Results	✓	UNITED only outperforms at highest shot budgets; individual methods can fail.

Table 50: Framework analysis: Bultrini et al. (2023).

6.2. Cirstoiu et al. (2023) — Volumetric Benchmarking of Error Mitigation with Qermit [46]

Volumetric benchmarking methodology for EM. Introduces Qermit open-source Python package. Finds disconnects between predicted and practical mitigation performance.

Criterion	Rating	Notes
Sample Size	✓	Volumetric benchmarks across width/depth parameter space.
Variance Reporting	✓	Performance distributions shown; uncertainty quantified.
Statistical Tests	△	Volumetric plots; no formal tests.
Drift Control	△	Hardware experiments; drift not explicitly controlled.
Overhead Accounting	✓	Overhead explicitly part of benchmarking framework.
Noise Model	✓	Real hardware; compares simulated vs. actual mitigation.
Reproducibility	✓	Qermit open-source package released; methodology specified.
Negative Results	✓	Predicted performance substantially overestimates actual performance.

Table 51: Framework analysis: Cirstoiu et al. (2023).

6.3. Prodius et al. (2025) — Robust Design Under Uncertainty in QEM [47]

General and unbiased methods for quantifying uncertainty of error-mitigated observables. Optimizes CDR training circuits and ZNE noise-level/shot allocation via surrogate-based optimization.

Criterion	Rating	Notes
Sample Size	✓	$N_{\text{tot}} = 10^4$ shots, $N_t = 10$ training circuits; explicitly specified.
Variance Reporting	✓	Central contribution: uncertainty quantification for mitigated observables.
Statistical Tests	△	Optimization validated empirically; no formal hypothesis tests.
Drift Control	n/a	Simulation-based study.

Overhead Accounting	✓	Shot allocation optimization accounts for overhead.
Noise Model	✓	IBM Toronto noise model + depolarizing; transferability tested.
Reproducibility	✓	Parameters fully specified; demonstrates transferable methodology.
Negative Results	△	Degradation under high noise; limited failure analysis.

Table 52: Framework analysis: Prodius et al. (2025).

6.4. Mamedaliev et al. (2026) — Framework to Evaluate VQA Performance [48]

Evaluation framework for VQAs on QUBO combining feasibility, quality, and reproducibility metrics. Reproducibility formalised via Shannon entropy. Proof-of-concept on 16-qubit QUBO.

Criterion	Rating	Notes
Sample Size	✓	400 VQA runs per configuration; varying shot counts.
Variance Reporting	✓	VQA distributions shown; Shannon entropy quantifies spread.
Statistical Tests	△	Custom quality function; no standard statistical tests.
Drift Control	n/a	Simulation-based.
Overhead Accounting	✓	Resource consumption central to quality diagram.
Noise Model	△	NISQ context; specific noise model not detailed.
Reproducibility	✓	Reproducibility is a core metric; methodology specified.
Negative Results	△	Certain VQA configurations shown infeasible.

Table 53: Framework analysis: Mamedaliev et al. (2026).

6.5. Moguel et al. (2024) — Quantum Software Experiments: Reporting Guidelines [49]

Proposes guidelines for reporting quantum experiments and laboratory package structures. Reviews 11 works for compliance with proposed criteria.

Criterion	Rating	Notes
Sample Size	n/a	Meta-methodology paper.
Variance Reporting	n/a	Evaluates others' reporting.
Statistical Tests	n/a	No original comparisons.
Drift Control	n/a	Not addressed directly.
Overhead Accounting	n/a	Not an experimental study.
Noise Model	n/a	Not addressed.
Reproducibility	✓	Central contribution: reproducibility guidelines for quantum experiments.
Negative Results	✓	Shows most reviewed works fail to meet reporting standards.

Table 54: Framework analysis: Moguel et al. (2024).

6.6. Zheng et al. (2023) — Bayesian Characterizing and Mitigating Gate/Measurement Errors [50]

Bayesian inference for posterior distributions of gate and measurement error parameters on IBM ibmqx2. Applied to state tomography, Grover's, QAOA, and 200-NOT-gate circuits.

Criterion	Rating	Notes
Sample Size	△	Multiple experiments; shot counts not always explicit.

Variance Reporting	✓	Bayesian posteriors provide full uncertainty quantification.
Statistical Tests	△	Bayesian model comparison; no frequentist tests.
Drift Control	△	Real hardware data but drift not explicitly controlled.
Overhead Accounting	△	Classical inference overhead acknowledged.
Noise Model	✓	Bit-flip gate/measurement error models validated on hardware.
Reproducibility	✓	Code referenced; parameters specified.
Negative Results	△	Limitations of simple noise models for complex circuits shown.

Table 55: Framework analysis: Zheng et al. (2023).

6.7. Pelofske et al. (2024) — Benchmarking QEM on Real Hardware

Comprehensive benchmarking study of QEM techniques across multiple IBM backends and circuit types. Evaluates ZNE, PEC, and readout error mitigation with varying circuit depths and qubit counts.

Criterion	**Rating**	**Notes**
Sample Size	✓	Multiple configurations with clearly stated shot counts.
Variance Reporting	✓	Statistical spreads across repetitions reported.
Statistical Evidence	△	Descriptive uncertainty reporting; no formal tests for method comparisons.
Drift Control	△	Partial hardware drift awareness.
Overhead Accounting	✓	Mitigation overhead quantified per method.
Noise Model	△	IBM backend noise used but limited independent characterisation.
Reproducibility	✓	Code and data publicly available.
Negative Results	✓	Failure modes at high circuit depths reported.

Table 56: Framework analysis: Pelofske et al. (2024).

7. Reproducibility & Drift

7.1. Hirasaki et al. (2023) — Detection of Temporal Fluctuation for QEM [51]

Detects temporal fluctuations in superconducting qubits via a fluctuation indicator S. Demonstrates QEM via post-selection, reducing Bell-state errors from 5.4% to 1.6%. Uses $L = 1,787,904$ measurements.

Criterion	Rating	Notes
Sample Size	✓	$L = 1,787,904$ total measurements; block parameters specified.
Variance Reporting	△	Monte Carlo validation; error bars in some figures but not systematic.
Statistical Tests	△	Fluctuation indicator S with threshold; no formal hypothesis tests.
Drift Control	✓	Central contribution: explicitly detects and addresses temporal drift.
Overhead Accounting	△	Post-selection reduces dataset size; overhead partially quantified.
Noise Model	△	Empirical noise characterisation; step-like model validated via Monte Carlo.
Reproducibility	△	Parameters specified; IBM cloud hardware; no code link.
Negative Results	△	QV circuit improvement modest (17.5%⟶12.0%); limitations acknowledged.

Table 57: Framework analysis: Hirasaki et al. (2023).

7.2. Senapati et al. (2023) — Redefining Reproducibility in Quantum Computing [52]

Studies reproducibility of QML on NISQ devices (IBM Manila, Belem, Oslo, Nairobi). Same model yields different results across days. Correlation between test accuracy and calibration features.

Criterion	Rating	Notes
Sample Size	△	Daily runs over weeks; exact shot/circuit counts not always explicit.
Variance Reporting	✓	Standard deviations; accuracy distributions across days.
Statistical Tests	△	Correlation analysis; regression but no formal significance tests.
Drift Control	✓	Core contribution: demonstrates and quantifies effect of calibration drift on QML.
Overhead Accounting	✗	Not addressed.
Noise Model	△	Real calibration data; noisy simulators shown unreliable.
Reproducibility	✓	Daily experimental data; methodology well-specified.
Negative Results	✓	Main finding: QML is not reproducible across days/devices; simulation fails.

Table 58: Framework analysis: Senapati et al. (2023).

7.3. Senapati et al. (2024) — PQML: Predictive Reproducibility on NISQ Machines [53]

Predicts QML outcomes across quantum computers via 15 regression methods + 3 data transformations. 12-month data collection. Random forest achieves 95-97.1% prediction accuracy.

Criterion	Rating	Notes
Sample Size	✓	12-month data collection; multiple machines tested daily.
Variance Reporting	✓	Prediction accuracy with standard deviations; cross-validation.
Statistical Tests	△	Compares 15 methods; ranking but no formal significance tests.

Drift Control	✓	Core: predicts drift effect on reproducibility using calibration data.
Overhead Accounting	✗	Not addressed.
Noise Model	△	Calibration features as proxy; not a physical noise model.
Reproducibility	✓	Methodology specified; tool described.
Negative Results	✓	Inherent unpredictability on certain devices documented.

Table 59: Framework analysis: Senapati et al. (2024).

8. Reviews & Surveys

8.1. Cai et al. (2023) — Quantum Error Mitigation (Rev. Mod. Phys.) [54]

Comprehensive review of QEM methods, theory, comparisons, combinations, and applications. Defines the formal MSE decomposition and sampling overhead framework.

Criterion	Rating	Notes
Sample Size	n/a	Review paper.
Variance Reporting	✓	Formally defines MSE = bias² + variance; sampling overhead framework.
Statistical Tests	n/a	Discusses estimation theory but does not perform tests.
Drift Control	△	Mentions noise drift as challenge; no systematic treatment.
Overhead Accounting	✓	Central theme; C_{em} formally defined; exponential scaling proven.
Noise Model	✓	Surveys noise models across all QEM methods comprehensively.
Reproducibility	n/a	References open-source tools (Mitiq, QREM).
Negative Results	✓	Extensively discusses limitations, failure conditions, open problems.

Table 60: Framework analysis: Cai et al. (2023).

8.2. Bharti (2022) — Noisy Intermediate-Scale Quantum Algorithms (Rev. Mod. Phys.) [55]

Major review (60+ pages) of NISQ algorithms: VQAs, quantum annealing, boson sampling, analog simulation, error mitigation, circuit compilation, and benchmarking.

Criterion	Rating	Notes
Sample Size	n/a	Review paper.
Variance Reporting	n/a	No original data.
Statistical Tests	n/a	Review.
Drift Control	n/a	Not addressed.
Overhead Accounting	△	Discusses QEM overhead qualitatively.
Noise Model	△	Reviews multiple noise models; no original validation.
Reproducibility	n/a	Review paper.
Negative Results	✓	Thoroughly discusses barren plateaus, trainability, noise sensitivity.

Table 61: Framework analysis: Bharti (2022).

8.3. Lorenz et al. (2025) — Systematic Benchmarking of Quantum Computers [56]

European consortium survey/recommendations for systematic benchmarking: component-level, system-level, software-level, HPC/cloud-level, and application-level benchmarks.

Criterion	Rating	Notes
Sample Size	n/a	Review/recommendations paper.
Variance Reporting	n/a	No original data.
Statistical Tests	n/a	No original comparisons.
Drift Control	△	Discusses drift as benchmarking concern.
Overhead Accounting	△	Discusses overhead as consideration.
Noise Model	△	Reviews noise models in benchmarking context.

Reproducibility	✓	Explicitly advocates for reproducibility standards.
Negative Results	△	Identifies gaps in current practices.

Table 62: Framework analysis: Lorenz et al. (2025).

8.4. Njoki (2024) — Statistical Challenges in Quantum Computing [57]

Brief (5-page) high-level review of statistical challenges in quantum computing: QEC, statistical inference, quantum data interpretation, algorithm optimization.

Criterion	Rating	Notes
Sample Size	n/a	Review.
Variance Reporting	n/a	No original data.
Statistical Tests	n/a	Abstract discussion.
Drift Control	n/a	Not addressed.
Overhead Accounting	n/a	Not addressed.
Noise Model	n/a	Mentions noise conceptually.
Reproducibility	✗	No code/data; very brief review.
Negative Results	✗	No original findings.

Table 63: Framework analysis: Njoki (2024).

8.5. Régent (2025) — Awesome Quantum Computing Experiments [58]

Reviews and benchmarks experimental FTQC progress across platforms. Exponential trend fitting to 25 years of data. Open-source databases and analysis code.

Criterion	Rating	Notes
Sample Size	n/a	Meta-analysis/survey.
Variance Reporting	△	Goodness-of-fit metrics; limited uncertainty on individual points.
Statistical Tests	△	Exponential fits with R^2; no formal hypothesis testing.
Drift Control	n/a	Not applicable.
Overhead Accounting	n/a	Not an EM study.
Noise Model	n/a	Reviews platform characteristics.
Reproducibility	✓	Open-source GitHub repository with databases and code.
Negative Results	△	Identifies persistent challenges toward scalable FTQC.

Table 64: Framework analysis: Régent (2025).

8.6. Khan (2025) — Quantum Software Engineering [59]

Overview/tutorial on Quantum Software Engineering for IBM Systems. Covers Qiskit, VQE, Grover's, QAOA, noise mitigation, DevOps.

Criterion	Rating	Notes
Sample Size	n/a	Tutorial/review.
Variance Reporting	n/a	No original data.
Statistical Tests	n/a	No comparisons.
Drift Control	n/a	Not addressed.
Overhead Accounting	n/a	Not addressed.
Noise Model	△	Discusses noise conceptually.

Reproducibility	✗	No code/data; pedagogical.
Negative Results	✗	No original findings.

Table 65: Framework analysis: Khan (2025).

8.7. Younis et al. (2024) — Quantum Computing for NP-Hard Problems [60]

Survey of quantum algorithms for NP-hard problems. Review paper without original experiments.

Criterion	Rating	Notes
Sample Size	n/a	Review.
Variance Reporting	n/a	No original data.
Statistical Tests	n/a	No comparisons.
Drift Control	n/a	Not addressed.
Overhead Accounting	n/a	Not addressed.
Noise Model	n/a	Mentions noise conceptually.
Reproducibility	✗	No code/data.
Negative Results	△	Discusses known algorithmic limitations.

Table 66: Framework analysis: Younis et al. (2024).

8.8. LaRose et al. (2022) — Mitiq: A Software Package for Error Mitigation on NISQ Computers [61]

Introduces Mitiq, the first open-source Python library for quantum error mitigation. Implements ZNE, PEC, and Clifford Data Regression (CDR) with a hardware-agnostic interface supporting Cirq, Qiskit, pyQuil, and Braket. Provides benchmarks demonstrating error reduction on simulated and IBM hardware (1-4 qubits). The paper's primary contribution is software-engineering: standardised QEM workflows with composable interfaces. Benchmarks show ZNE typically reduces error by 2-5× with modest overhead for shallow circuits.

Criterion	Rating	Notes
Sample Size	△	Benchmark circuits tested with 10^3-10^4 shots; limited number of problem instances per method.
Variance Reporting	✓	Error bars via repeated runs shown in benchmark plots. Discusses variance amplification from ZNE extrapolation order.
Statistical Tests	✗	No formal tests. Method comparisons via error reduction plots.
Drift Control	✗	No drift discussion. Cloud hardware benchmarks without temporal controls.
Overhead Accounting	✓	Shot overhead from noise amplification quantified. PEC overhead γ discussed. Method-specific costs documented.
Noise Model	△	Uses default Qiskit/Cirq noise models for simulated benchmarks. Hardware noise not independently characterised.
Reproducibility	✓	Mitiq is fully open source (GitHub). All benchmarks reproducible via library API. Pip-installable.
Negative Results	✓	Documents cases where higher-order ZNE increases error. Notes PEC overhead can be prohibitive for deep circuits. Shows CDR training set size limitations.

Table 67: Framework analysis: LaRose et al. (2022) — Mitiq.

8.9. AbuGhanem (2024) — A Decade of Google Quantum AI [62]

Narrative review of Google Quantum AI's trajectory from Sycamore random circuit sampling (2019) through Willow (2024). Covers quantum supremacy, error correction (surface code below threshold), and quantum chemistry applications. Not a systematic review — provides context and historical perspective rather than original analysis.

Criterion	**Rating**	**Notes**
Sample Size	n/a	Historical review; no original experiments.
Variance Reporting	△	Cites original papers' error bars; no independent analysis.
Statistical Tests	n/a	No original comparisons.
Drift Control	n/a	Not addressed.
Overhead Accounting	△	Discusses overhead conceptually in the context of error correction vs. mitigation trade-offs.
Noise Model	△	Discusses noise models used by Google (depolarising, Pauli) at a high level without technical depth.
Reproducibility	△	Comprehensive citation list; no original methodology to reproduce.
Negative Results	✗	Largely celebratory narrative. Does not critically evaluate limitations or failures.

Table 68: Framework analysis: AbuGhanem (2024).

8.10. Huang et al. (2023) — Near-Term Quantum Computing Techniques: VQA, QEM, QECC

Comprehensive survey of near-term quantum computing techniques covering variational quantum algorithms, quantum error mitigation, and quantum error-correcting codes. Provides a taxonomy of approaches and discusses connections between VQA, QEM, and QECC paradigms.

Criterion	**Rating**	**Notes**
Sample Size	n/a	Survey paper; no original experiments.
Variance Reporting	✓	Reviews variance reporting practices across surveyed papers.
Statistical Evidence	△	Discusses statistical methodology conceptually but no original analysis.
Drift Control	△	Drift mentioned as a challenge but not analysed.
Overhead Accounting	✓	Overhead concepts well covered in survey.
Noise Model	✓	Comprehensive noise model taxonomy.
Reproducibility	△	Survey format; references primary sources.
Negative Results	✓	Discusses limitations of each paradigm.

Table 69: Framework analysis: Huang et al. (2023).

9. Adjacent Topics

9.1. Hilder et al. (2022) — Fault-Tolerant Parity Readout on Trapped Ions [63]

Fault-tolerant weight-4 parity check on shuttling-based trapped-ion processor. Parity fidelity 92.3(2)%, improving to 93.2(2)% with flag conditioning.

Criterion	Rating	Notes
Sample Size	✓	960 shots per input state; multiple input states.
Variance Reporting	✓	Statistical uncertainties as parenthetical errors, e.g. 92.3(2)%.
Statistical Tests	△	Fidelity comparisons; no formal hypothesis testing.
Drift Control	△	Shuttling architecture inherently controls some drift.
Overhead Accounting	n/a	QEC experiment.
Noise Model	✓	Cycle benchmarking for two-qubit fidelities 99.6(2)%.
Reproducibility	△	Parameters specified; no open-source code/data.
Negative Results	△	Shows limitations of non-fault-tolerant schemes.

Table 70: Framework analysis: Hilder et al. (2022).

9.2. Noel et al. (2022) — Measurement-Induced Quantum Phases [64]

Observes measurement-induced phase transitions on trapped-ion quantum computer (up to 13 qubits). 4000 shots $\times$ 3 bases $\times$ 300 circuit instances.

Criterion	Rating	Notes
Sample Size	✓	4000 shots $\times$ 3 bases $\times$ 300 circuits; clearly specified.
Variance Reporting	✓	1σ uncertainty; error bars in all figures.
Statistical Tests	△	Phase transitions via order parameters; no formal tests.
Drift Control	△	Gate fidelities reported; drift not explicitly addressed.
Overhead Accounting	n/a	Not an error mitigation study.
Noise Model	△	Gate fidelities via randomized benchmarking; no full noise model.
Reproducibility	△	Parameters specified; no code/data link.
Negative Results	✓	Finite system size limitations; post-selection overhead.

Table 71: Framework analysis: Noel et al. (2022).

9.3. Wu et al. (2022) — Erasure Conversion for Fault-Tolerant QC [65]

Erasure conversion for FTQC using ^{171}Yb neutral atoms. 98% of errors convertible to erasures. Surface code threshold increase from 0.937% to 4.15%.

Criterion	Rating	Notes
Sample Size	✓	Monte Carlo simulations for threshold estimation.
Variance Reporting	△	Threshold values with some uncertainty; not always explicit error bars.
Statistical Tests	✗	Threshold crossings identified visually.
Drift Control	n/a	Simulation/theoretical.
Overhead Accounting	△	Erasure conversion overhead discussed qualitatively.
Noise Model	✓	Detailed ^{171}Yb noise model; erasure fraction rigorously derived.
Reproducibility	△	Simulation methodology described; no code.

Negative Results	△	Acknowledges residual non-erasure errors.

Table 72: Framework analysis: Wu et al. (2022).

9.4. English et al. (2025) — Ising on the Donut: Topological QEC from Statistical Mechanics [66]

Analytic expressions for logical failure rates of post-selected toric code under bit-flip noise via exact mapping to 2D Ising model. Four distinct QEC regimes identified.

Criterion	Rating	Notes
Sample Size	n/a	Analytical/theoretical.
Variance Reporting	n/a	Exact analytic results.
Statistical Tests	n/a	No empirical comparisons.
Drift Control	n/a	Theoretical.
Overhead Accounting	△	Code distance vs. failure rate scaling; overhead implicit.
Noise Model	✓	Bit-flip noise exactly mapped to Ising model.
Reproducibility	△	Mathematical framework specified; no code.
Negative Results	✓	Characterises above-threshold regime where QEC fails.

Table 73: Framework analysis: English et al. (2025).

9.5. Chernyak et al. (2026) — Classical Regularization in VQE [67]

L2 regularization for stabilizing VQE optimization. Tested on H_2 (4-qubit), LiH (8-qubit), RFIM (12-qubit). Thousands of random initializations. Noiseless simulations.

Criterion	Rating	Notes
Sample Size	✓	Thousands of random initializations; multiple molecular/spin systems.
Variance Reporting	✓	Distribution of converged energies; histograms over initializations.
Statistical Tests	△	Visual distribution comparisons; no formal tests.
Drift Control	n/a	Noiseless simulations.
Overhead Accounting	△	Iteration counts reported; regularization overhead not quantified.
Noise Model	✗	Noiseless only.
Reproducibility	✓	Parameters specified; Qiskit implementation detailed.
Negative Results	✓	Regularization can worsen results if hyperparameters poorly chosen.

Table 74: Framework analysis: Chernyak et al. (2026).

9.6. Etim & Szefer (2025) — Fault Injection Attacks on ML-based QC Readout Error Correction [68]

First physical fault-injection study on ML-based quantum readout correction via voltage glitching. Layer-dependent susceptibility characterised via Hamming distance analysis.

Criterion	Rating	Notes
Sample Size	✓	96 fault attempts per layer per configuration; systematic sweep.
Variance Reporting	△	Fault rates reported; limited error bars.
Statistical Tests	△	Hamming distance analysis; no formal hypothesis tests.
Drift Control	n/a	Controlled fault injection, not quantum drift.
Overhead Accounting	n/a	Security study.

Noise Model	△	Voltage glitching model characterised empirically.
Reproducibility	✓	ChipWhisperer setup and parameters fully specified.
Negative Results	✓	Certain layers highly susceptible to attacks.

Table 75: Framework analysis: Etim & Szefer (2025).

9.7. Acharya et al. (2025) — Quantum Error Correction Below the Surface Code Threshold [69]

Google's Willow chip (105 transmon qubits) demonstrating QEC below threshold for the first time with the surface code. Error suppression factor $\Lambda = 2.14 \pm 0.02$ per code distance step (from $d = 3$ to $d = 7$), meaning logical error rate decreases exponentially with code size. Real-time decoding at 63 μs latency. While this is a QEC (not QEM) paper, it is critical context: it demonstrates the hardware trajectory toward fault tolerance and establishes the baseline against which QEM's "bridge" role must be evaluated.

Criterion	Rating	Notes
Sample Size	✓	10^4-10^6 syndrome measurement rounds per experiment; 16 stabiliser qubits for $d = 3$, 72 for $d = 7$. Multiple code distances. Extensive statistics.
Variance Reporting	✓	Error bars from binomial statistics on logical error rates. Confidence intervals on Λ from bootstrap. Distribution of detector error rates shown.
Statistical Tests	✓	Threshold crossing established via statistical fit of $\Lambda > 1$ with confidence interval. Leakage and detection event correlations tested.
Drift Control	✓	Leakage removal circuits (MLR) applied every round. Real-time calibration loop. Experiments designed around calibration cadence.
Overhead Accounting	✓	Qubit overhead (d^2 data + $(d-1)^2 + d^2$ ancilla for each logical qubit) explicit. Space-time overhead of QEC reported.
Noise Model	✓	Stochastic Pauli noise with correlated errors. Detector error model validated against hardware via Monte Carlo simulations. Leakage modelled and mitigated.
Reproducibility	✓	Full device specifications, calibration procedures, and decoder details published. Supplementary data available.
Negative Results	✓	Reports residual correlated errors limiting Λ. Shows leakage accumulation without active removal. Acknowledges gap between physical and logical error rates.

Table 76: Framework analysis: Acharya et al. (2025) — Google Willow.

9.8. Sahu & Gupta (2024) — NAC-QFL: Noise-Aware Clustered Quantum Federated Learning [70]

Proposes noise-aware clustered quantum federated learning (NAC-QFL) for distributed quantum machine learning. Uses circuit cutting to partition circuits across heterogeneous noisy backends, then aggregates results with noise-aware weighting. The noise-awareness is the relevant QEM angle: the framework clusters devices by noise profile and weights contributions to mitigate the effect of noisy backends on the aggregated model.

Criterion	Rating	Notes
Sample Size	△	Multiple datasets (MNIST, Fashion-MNIST); limited number of federated rounds reported. Backend count varies (2-4 devices).

Variance Reporting	✓	Training/test accuracy curves with error bands across federated rounds. Inter-client variance reported.
Statistical Tests	✗	No formal tests. Accuracy comparisons via tables and plots.
Drift Control	n/a	Simulated noise profiles; no real hardware drift.
Overhead Accounting	△	Circuit cutting overhead discussed qualitatively. Communication rounds reported. Classical post-processing cost not quantified.
Noise Model	✓	Noise-aware clustering uses per-device noise profiles (gate errors, readout errors, T_1/T_2). Noise strength varied systematically.
Reproducibility	△	Framework described in detail. No public code repository. Datasets standard (MNIST variants).
Negative Results	△	Shows performance degrades with increasing noise heterogeneity. Clustering can fail when all devices are equally noisy.

Table 77: Framework analysis: Sahu & Gupta (2024).

9.9. Tindall et al. (2024) — Efficient Tensor Network Simulation of IBM's Eagle Kicked Ising Experiment [71]

Demonstrates that the 127-qubit kicked Ising experiment from Kim et al. (2023) — which used ZNE to claim quantum utility — can be classically simulated to **much higher** accuracy using belief-propagation tensor networks (BP-TNS) reflecting the heavy-hexagon lattice geometry. The method achieves $\sim 10^{-14}$ accuracy for 5-step magnetisation on a laptop in under 10 seconds, vs. the quantum processor's $\sim 10^{-2}$ errors. Extends to the thermodynamic (infinite-qubit) limit. This is the canonical classical spoiler paper that critically undermines the statistical meaning of Kim et al.'s ZNE-mitigated results and illustrates how model-specific structure can be exploited classically.

Criterion	Rating	Notes
Sample Size	△	Classical TN simulations; convergence studied via $1/\chi$ extrapolation (bond dimension up to $\chi = 500$). Deterministic — no shot-based sampling.
Variance Reporting	△	Error quantified by absolute difference from exact results and BP separability metric. Extrapolation uncertainty shown as shaded regions (Fig. 4b). No statistical error bars per se.
Statistical Tests	✗	No formal tests. Comparisons via absolute error on log scale against quantum processor and other classical methods.
Drift Control	n/a	Purely classical computation. However, the paper's existence questions whether Kim et al.'s drift-unaware QEM results were reliable.
Overhead Accounting	△	Wall time (seconds on M1 laptop) and memory reported. Not compared to quantum sampling overhead in a unified framework.
Noise Model	n/a	Classical noiseless simulation. Indirectly demonstrates that QEM-mitigated quantum results can be bettered without any noise model.
Reproducibility	✓	Code: ITensorNetworks.jl (open-source Julia). Data: github.com/JoeyT1994/BP-TNS-Data. Parameters fully specified.
Negative Results	△	Acknowledges BP approximation degrades for larger θ_h and deeper circuits. Shows MPS error grows for $\theta_h = 1.0$ beyond ~ 10 steps. Does not claim universality.

Table 78: Framework analysis: Tindall et al. (2024).

9.10. Campbell et al. (2023) — Superstaq: Deep Optimization of Quantum Programs [72]

Describes Superstaq, a cross-layer quantum compilation platform that optimises circuits to device-native gate sets across IBM, Rigetti, QSCOUT (trapped ion), AQT@LBNL (superconducting), and Infleqtion (neutral atom) hardware. Achieves $2.3\times$–$13\times$ Hellinger fidelity improvements for QCC chemistry circuits and up to $68\times$ relative-strength improvements for Bernstein-Vazirani with dynamical decoupling. While not a QEM paper per se, it integrates dynamical decoupling and error-aware compilation techniques and provides multi-platform hardware benchmarks relevant to understanding pre-mitigation circuit quality — the "input quality" that QEM methods operate on.

Criterion	**Rating**	**Notes**
Sample Size	△	Shot counts specified for some experiments (32k for QCC, 4000 for BV) but not systematically for all. 10 trials for QAOA, 50 FSim instances for QSCOUT.
Variance Reporting	△	Error bars shown for QAOA (standard error of mean) and QSCOUT FSim (2σ bars). Most other results report only point estimates or single-run Hellinger fidelities.
Statistical Tests	✗	No formal hypothesis tests. Comparisons by Hellinger fidelity ratios and relative strength metrics. QSCOUT linear fit with 2σ bands is the closest to formal analysis.
Drift Control	✗	No discussion of temporal drift. The BYOG paradigm implicitly handles day-to-day gate variation via re-characterisation but is not framed as drift control.
Overhead Accounting	△	Circuit depth/duration savings quantified (e.g., 160dt saved per CX, 42% pulse reduction). No systematic sampling overhead analysis. DD overhead not accounted against decoherence benefit.
Noise Model	△	Depolarising model used for neutral atom fidelity estimation. Authors explicitly caveat: "this agreement should not be taken too seriously." No systematic noise model validation.
Reproducibility	✓	Open-source clients `cirq-superstaq` and `qiskit-superstaq` on PyPI. Hardware targets and gate decompositions specified. Cloud compilation service itself is proprietary.
Negative Results	△	Notes that direct AceCR implementation underperforms Qiskit for some configurations. Acknowledges loss of expressivity in BYOG approach. Does not systematically report failure cases.

Table 79: Framework analysis: Campbell et al. (2023) — Superstaq.

10. Global Analysis: Summary Table

Paper	1	2	3	4	5	6	7	8
QEM Method Papers								
Tran et al. 2023	△	✓	✗	n/a	✓	✓	△	△
Filippov et al. 2024	✓	✓	✗	✓	✓	✓	△	✓
Khan et al. 2024	△	△	✗	✗	✗	△	△	△
McDonough et al. 2022	△	△	✗	✗	✓	✓	✓	△
Zhang et al. 2025	✓	✓	△	△	✓	✓	✓	✓
Qin et al. 2023	✓	✓	✗	n/a	✓	✓	△	✓
Liao et al. 2024	✓	✓	△	△	✓	✓	✓	✓
Majumdar et al. 2023	✓	✓	✗	△	✓	✓	✓	✓
Wahl et al. 2023	✓	△	✗	n/a	✓	△	✓	△
Hour et al. 2024	△	△	✗	✗	△	✓	△	△
Jnane et al. 2024	△	✓	n/a	△	✓	✓	△	△
Henao et al. 2023	△	△	✗	✓	△	✓	△	✓
Filippov et al. 2023	✓	✓	✗	n/a	✓	✓	△	△
LaRose et al. 2022	✓	✓	△	✗	✓	△	✓	△
Pascuzzi et al. 2022	✓	✓	✗	△	△	✓	✓	△
Liu & Cai 2025	✓	✓	△	n/a	✓	△	△	△
Seif et al. 2023	✓	✓	△	△	✓	△	△	✓
Hicks et al. 2022	△	△	✗	✗	✓	✓	△	✓
van den Berg et al. 2022	✓	✓	△	△	✓	✓	✓	△
Krebsbach et al. 2022	✓	✓	✗	n/a	✓	△	✓	✓
Kim et al. 2025 (enhanced)	✓	△	✗	✗	△	△	△	✓
Kiss et al. 2025	△	✓	△	△	△	✓	△	△
Jin et al. 2025	n/a	n/a	n/a	n/a	✓	△	△	△
van den Berg et al. 2023	✓	✓	△	△	✓	✓	✓	△
Hosseinkhani et al. 2025	✓	✓	△	✗	✓	△	△	△
Fundamental Limits & Theory								
Takagi et al. 2022	n/a	n/a	n/a	n/a	✓	✓	✓	✓
Takagi et al. 2023	✓	✓	n/a	n/a	✓	✓	△	✓
Quek et al. 2024	n/a	n/a	n/a	n/a	✓	✓	✓	✓
Niroula et al. 2025	△	△	✗	n/a	✓	✓	△	✓
Jose & Simeone 2022	✓	✓	✗	n/a	✓	△	△	✓
White 2025	△	✓	✗	✓	✓	✓	✓	✓
França & Garcia-Patrón 2022	n/a	n/a	n/a	n/a	✓	✓	△	✓
Experimental Demonstrations								
Saki et al. 2023	✓	✓	✓	△	✓	△	✓	✓
Govia et al. 2025	△	△	△	✓	△	✓	✓	✓

Maupin et al. 2024	✓	✓	△	△	✓	✓	✓	✓
Kurita et al. 2023	△	✓	✗	n/a	△	✓	✓	✓
Russo et al. 2023	✓	✓	△	✗	✓	△	✓	✓
Kim et al. 2025 (stabilized)	✓	✓	△	✓	✓	✓	✓	✓
Weaving et al. 2023	✓	✓	△	△	△	✓	✓	✓
Dutkiewicz et al. 2025	✓	✓	△	n/a	✓	✓	✓	✓
Amin et al. 2023	△	△	✗	✓	△	✓	△	✓
Kim et al. 2023 (utility)	✓	✓	△	△	✓	✓	✓	✓
Roy et al. 2024	△	✗	✗	✗	△	△	✓	✓
Shi & Malaney 2023	✓	△	✗	✗	✓	△	△	✓
Venere et al. 2024	△	△	✗	n/a	✗	△	△	△
Benchmarking & Methodology								
Bultrini et al. 2023	✓	✓	△	n/a	✓	✓	△	✓
Cirstoiu et al. 2023	✓	✓	△	△	✓	✓	✓	✓
Prodius et al. 2025	✓	✓	△	n/a	✓	✓	✓	△
Mamedaliev et al. 2026	✓	✓	△	n/a	✓	△	✓	△
Moguel et al. 2024	n/a	n/a	n/a	n/a	n/a	n/a	✓	✓
Zheng et al. 2023	△	✓	△	△	△	✓	✓	△
Anand et al. 2023	△	△	✗	n/a	△	△	△	✓
Desdentado et al. 2025	△	✗	✗	✗	△	△	✓	△
Reproducibility & Drift								
Hirasaki et al. 2023	✓	△	△	✓	△	△	△	△
Senapati et al. 2023	△	✓	△	✓	✗	△	✓	✓
Senapati et al. 2024	✓	✓	△	✓	✗	△	✓	✓
Reviews & Surveys								
Cai et al. 2023	n/a	✓	n/a	△	✓	✓	n/a	✓
Bharti 2022	n/a	n/a	n/a	n/a	△	△	n/a	✓
Lorenz et al. 2025	n/a	n/a	n/a	△	△	△	✓	△
Njoki 2024	n/a	n/a	n/a	n/a	n/a	n/a	✗	✗
Régent 2025	n/a	△	△	n/a	n/a	n/a	✓	△
Khan 2025	n/a	n/a	n/a	n/a	n/a	△	✗	✗
Younis et al. 2024	n/a	n/a	n/a	n/a	n/a	n/a	✗	△
LaRose et al. 2022 (Mitiq)	△	✓	✗	✗	✓	△	✓	✓
AbuGhanem 2024	n/a	△	n/a	n/a	△	△	△	✗
Adjacent Topics								
Hilder et al. 2022	✓	✓	△	△	n/a	✓	△	△
Noel et al. 2022	✓	✓	△	△	n/a	△	△	✓
Wu et al. 2022	✓	△	✗	n/a	△	✓	△	△
English et al. 2025	n/a	n/a	n/a	n/a	△	✓	△	✓

Chernyak et al. 2026	✓	✓	△	n/a	△	✗	✓	✓
Etim & Szefer 2025	✓	△	△	n/a	n/a	△	✓	✓
Acharya et al. 2025 (Willow)	✓	✓	✓	✓	✓	✓	✓	✓
Sahu & Gupta 2024	△	✓	✗	n/a	△	✓	△	△
Tindall et al. 2024	△	△	✗	n/a	△	n/a	✓	△
Campbell et al. 2023	△	△	✗	✗	△	△	✓	△
Foundational Papers (Pre-2022)								
Temme et al. 2017	△	△	✗	n/a	✓	✓	△	✓
Li & Benjamin 2017	△	✓	✗	✗	✓	✓	△	✓
Endo et al. 2018	✓	✓	✗	✗	✓	✓	△	✓
Kandala et al. 2019	✓	✓	✗	△	△	△	△	✓
Czarnik et al. 2021	✓	✓	✗	✗	✓	△	△	✓
Wang et al. 2021	✓	✓	△	△	✓	✓	△	✓
Urbanek et al. 2021	✓	✓	✗	✗	△	✓	△	△
Strikis et al. 2021	✓	✓	✗	✓	✓	✓	△	✓

Table 80: Global summary table. Columns: 1=Sample Size, 2=Variance, 3=Stat. Tests, 4=Drift, 5=Overhead, 6=Noise Model, 7=Reproducibility, 8=Neg. Results.

11. Foundational Papers (2021)

These early papers established key QEM techniques and are included as historical baselines for the field's statistical practices.

11.1. Czarnik et al. (2021) — Error Mitigation with Clifford Quantum-Circuit Data [73]

Introduces Clifford Data Regression (CDR), a learning-based QEM method that trains a linear model on paired noisy/exact data from near-Clifford circuits (efficiently classically simulable) and applies it to correct observables from arbitrary circuits. Demonstrates order-of-magnitude error reduction for QAOA ground-state energy estimation on 16 qubits on IBM Almaden and 64 qubits on a noisy simulator. The method's shot cost scales polynomially with qubit count, and it outperforms ZNE for larger circuits.

Criterion	Rating	Notes
Sample Size	✓	16384 shots per circuit; 60-80 training circuits per target state; total mitigation cost $\approx 10^6$ shots. Convergence studied as function of training set size N (Fig. 3a). System sizes $Q = 8$-64.
Variance Reporting	✓	Error bars derived from training cost function C (3 standard deviations). Mean and maximum relative errors plotted across 30 optimisation instances. Convergence with shot count shown.
Statistical Tests	✗	No formal hypothesis tests. Comparisons via mean/max relative error plots and tables. No p-values.
Drift Control	✗	No discussion of temporal drift. IBM Almaden data collected without interleaving or drift-correction protocol.
Overhead Accounting	✓	Total shot cost of mitigation explicitly reported (1.05×10^6 for 16-qubit hardware, 1.16×10^6 for simulator). Training cost scaling analysed.
Noise Model	△	Uses GST-derived noise model from IBM Ourense for simulator experiments. Acknowledges it "does not capture spatial and time inhomogeneous aspects of the real device noise" and omits measurement errors in simulation.
Reproducibility	△	Circuit structure, QAOA parameters, noise model process matrices referenced. No code repository (method is straightforward to reimplement). IBM Almaden device since retired.
Negative Results	✓	Reports that CDR performance depends on number of non-Clifford gates N (Fig. 3a shows degradation for too few or too many). Notes noise model limitations. Compares unfavourably at very small circuit sizes where ZNE suffices. Shows CDR breaks down for circuits with very deep non-Clifford content.

Table 81: Framework analysis: Czarnik et al. (2021).

11.2. Wang et al. (2021) — Noise-Induced Barren Plateaus in Variational Quantum Algorithms [74]

Proves that for variational quantum algorithms under local Pauli noise, the gradient of the cost function vanishes exponentially in circuit depth (and system size for global cost functions): $\langle \partial_\theta C \rangle \sim e^{-\alpha n d}$ where α depends on the noise rate. This "noise-induced barren plateau" (NIBP) is distinct from (and compounds) the entanglement-induced barren plateaus known for deep random circuits. The result implies that for any fixed noise rate $p > 0$, there exists a critical depth d^* beyond which variational quantum algorithms cannot be trained — even with QEM, since mitigating the noise requires exponential overhead that cancels any gradient signal recovery. This paper provides a fundamental constraint on the utility of near-term variational approaches and directly limits the regime where QEM can be productive.

Criterion	Rating	Notes
Sample Size	✓	Numerical simulations sweep system size $n = 4$-16 qubits and depth $d = 1$-100. Gradient computed via parameter-shift rule with 10^3-10^4 shots per evaluation. Multiple random instances.
Variance Reporting	✓	Variance of gradient $\mathrm{Var}[\partial_\theta C]$ is the central observable. Analytical bounds on variance derived. Numerical variance verified against theory.
Statistical Tests	△	Analytical predictions compared with numerical estimates visually. No formal hypothesis tests, but tight agreement between theory and numerics demonstrated.
Drift Control	△	Simulated noise; no hardware drift. Noise parameters fixed throughout each experiment.
Overhead Accounting	✓	Proves that shot overhead to resolve gradient scales as $e^{\alpha n d}$, making training infeasible beyond d^*. Overhead scaling is the central result.
Noise Model	✓	Local depolarising and local Pauli channels. Generalisation to amplitude damping discussed. Analytical results general for any single-qubit Pauli channel.
Reproducibility	△	Full proofs and simulation parameters in appendix. No code repository, but simulation setup is standard (parameter-shift gradients with Pauli noise).
Negative Results	✓	The paper IS a negative result: proves noise makes VQA training exponentially hard. Shows no circuit architecture can avoid NIBPs under noise. Implies QEM has bounded utility for deep VQAs.

Table 82: Framework analysis: Wang et al. (2021).

11.3. Urbanek et al. (2021) — Mitigating Depolarizing Noise on Quantum Computers with Noise-Estimation Circuits [75]

Introduces noise-estimation circuits (NECs) that estimate the effective depolarising noise rate p by stripping single-qubit gates from a target circuit to create a companion circuit whose ideal output is known. The estimated p corrects the target circuit's expectation values via $\langle O \rangle = \overline{\langle O \rangle}/(1-p)$. Combined with readout-error correction, randomised compiling (448 instances), and ZNE ($1\times$, $3\times$, $5\times$ CNOT replacement with quadratic extrapolation), the method produces near-exact results for Heisenberg model simulation on IBM Q Paris with circuits containing up to 210 CNOT gates. The analytical argument for why NEC+ZNE outperforms ZNE alone is clean and instructive.

Criterion	Rating	Notes
Sample Size	✓	8192 shots per circuit; 448 randomised compiling instances per configuration. 6 qubits on IBM Q Paris. Circuit depth varies from 14 to 210 CNOTs (1-15 time steps).
Variance Reporting	✓	Error bars on all data figures: standard error of the mean for estimation circuits, standard deviation for mitigated results. Clearly distinguishes SEM vs. SD.
Statistical Tests	✗	No formal statistical tests or p-values. Comparisons are visual — mitigated vs. exact curves with error bars.

Drift Control	✗	No discussion of temporal drift. Point-in-time experiments on IBM Q Paris without interleaving or recalibration.
Overhead Accounting	△	3 circuit variants ($1\times$, $3\times$, $5\times$ CNOTs) plus estimation circuits. Total budget implicit ($448 \times 3 \times 2$ circuits) but overhead not explicitly quantified as a multiplicative factor.
Noise Model	✓	Depolarising noise model clearly stated. Randomised compiling explicitly converts coherent errors to incoherent (depolarising). Analytical justification for NEC+ZNE synergy derived.
Reproducibility	△	IBM Q Paris hardware, specific qubits (Q23-Q20), shot count, randomisation protocol fully specified. No code or data repository. Circuit diagrams provided.
Negative Results	△	Notes exponential extrapolation is "very sensitive to noise and leads to worse results." Acknowledges method requires CNOT gates to be the dominant error source.

Table 83: Framework analysis: Urbanek et al. (2021).

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